

Buffer Sizing and Bufferbloat in Routers

1 How Much Buffering Is Enough?

When bursts of packets arrive faster than the router can forward them, queues start building up. A fundamental question is:

How much buffering should a router have?

The answer depends heavily on TCP congestion control dynamics, traffic variance, and the number of flows sharing the bottleneck.

1.1 Classical Rule-of-Thumb (RFC 3439)

The long-standing classical rule is:

$$B = \text{RTT} \times C$$

Where:

- B = required buffer size,
- RTT = average round-trip time,
- C = link capacity.

$$\text{RTT} = 250 \text{ ms}, \quad C = 10 \text{ Gbps}$$

$$B = 2.5 \text{ Gbits}$$

This was designed when links carried **few TCP flows**.

1.2 Modern High-Capacity Links: Square-Root Rule

More recent work shows that when many TCP flows share the same link, buffering needs can drop significantly:

$$B = \frac{\text{RTT} \times C}{\sqrt{N}}$$

Where N = number of TCP flows.

This is especially relevant in backbone networks with thousands of flows.

1.3 Why Not Use Very Large Buffers?

While it seems like:

More buffer = fewer drops = better throughput

this is misleading.

Large buffers increase queueing delay, slow TCP reaction, and produce lag spikes even when the link is lightly loaded.

Interactive applications (gaming, voice, video) suffer heavily.

1.4 Bufferbloat: When Delay Explodes

Bufferbloat refers to **extreme latency caused by oversized buffers** that stay persistently full.

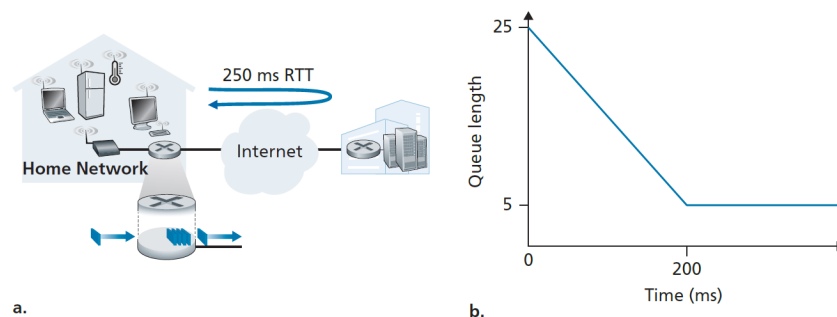


Figure 1: Bufferbloat in a home router.

1.5 Example: Home Router Bufferbloat

Assume:

- Transmission per packet = 20 ms
- Path RTT = 200 ms

A burst of 25 packets arrives at $t = 0$. Only one packet transmits every 20 ms.

The queue stabilizes at 5 packets:

$$\text{Queueing delay} = 5 \times 20 \text{ ms} = 100 \text{ ms}$$

even with no cross traffic.

This explains why home users feel lag even with low traffic.

1.6 Fixing Bufferbloat: AQM

Modern routers use AQM (Active Queue Management) to actively control latency:

- CoDel,
- PIE (RFC 8033),
- DOCSIS 3.1 AQM (RFC 8034).

AQM detects persistent queues and marks/drops packets to prevent bloating.

Conclusion

Buffer sizing is about balancing:

- throughput,
- delay,
- number of flows,
- TCP behavior.

Too little buffer $\rightarrow$ packet drops. Too much buffer $\rightarrow$ high latency (bufferbloat).
Modern networks use **AQM** to maintain low delay and high throughput.